

SK-Adapter

Skeleton-Based Structural Control for Native 3D Generation

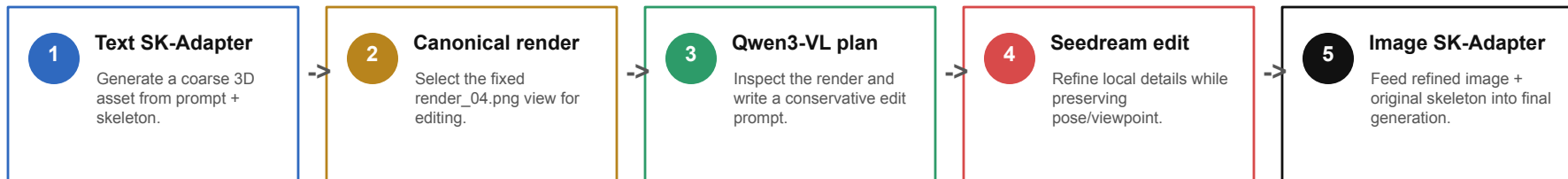
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What problem are we trying to solve?

- Skeleton control can preserve pose and broad structure, but text-only generation may miss prompt-specific visual details.
- Image-conditioned generation provides appearance guidance, yet the conditioning image can still be weak or ambiguous.
- The refinement target is not a new pose: improve semantic fidelity while keeping camera, silhouette, pose, and skeleton alignment stable.

Project question: can a single canonical render be edited into a better image-to-3D condition without breaking skeleton compatibility?

How does SK-Adapter++ refinement work?



Core constraint

The 2D editor is allowed to clarify texture, materials, facial markings, joints, and local shape cues. It should not rotate, mirror, re-pose, crop, or replace the object.

OpenRouter models: Qwen3-VL for planning; Seedream 4.5 for image editing.

Evaluation setup

Pipelines compared



Dataset note: refine aggregate block reports 50 samples.

Metrics

- CLIP mean view: text-image alignment over rendered views.
- PickScore mean view: preference-style image/text quality signal.
- DINO/KD: visual similarity signal; refine run has incomplete/buggy logging, so treat as cautionary.
- Anymate Chamfer: skeleton/rig alignment proxy; refine is currently N/A.

Quantitative evaluation

Reported metrics by pipeline; arrows indicate preferred direction

n = 50 for the Refine aggregate block

Pipeline	CLIP $\uparrow$ mean $\pm$ std	PickScore $\uparrow$ mean $\pm$ std	DINO/KD $\uparrow$ mean $\pm$ std	Anymate CD $\downarrow$ mean $\pm$ std	Note
Image	22.53 $\pm$ 3.90	19.97 $\pm$ 0.92	2.450 $\pm$ 0.200	0.738 $\pm$ 0.223	baseline
Text	26.37 $\pm$ 2.90	20.84 $\pm$ 0.97	2.766 $\pm$ 0.214	0.735 $\pm$ 0.219	baseline
Refine	26.87	21.10	2.618 $\uparrow$	N/A	proposed

CLIP comparison



PickScore comparison



Key observation Refine improves CLIP (+19.3% vs Image, +1.9% vs Text) and PickScore (+5.7% vs Image, +1.3% vs Text).

$\uparrow$ DINO/KD for Refine is retained as an incomplete-run indicator; Refine Anymate CD is not yet available.

Qualitative example: prompt-specific details become clearer



Image pipeline



Text pipeline



Refine pipeline

Prompt: compact futuristic robotic vehicle/drone, white-red scheme, cylindrical thrusters, articulated mechanical limbs, antennae, and shark-face teeth.

Qwen edit focus: sharpen teeth, clarify segmented joints, and emphasize cylindrical thrusters while preserving exact camera, pose, silhouette, and background.

Conclusion & takeaways

- SK-Adapter++ adds a lightweight refinement loop between coarse text+skeleton generation and final image+skeleton generation.
- The refine pipeline improves the main text-image quality metrics: CLIP and PickScore are highest among the tested pipelines.
- Qualitative evidence shows the refinement is most useful for local semantic details: teeth, joints, thruster edges, and material clarity.
- Current limitations: DINO/KD needs a confirmed refine rerun, and Refine Anymate Chamfer is not yet available.

Main takeaway: use 2D refinement as a conservative semantic correction step, not as a replacement for skeleton control.